

Part A: Student–Enrollment Relationship

Q1

```
SQL> SELECT
  2   S.StudentID,
  3   S.Name,
  4   E.CourseID
  5 FROM Student S
  6 INNER JOIN Enrollment E
  7 ON S.StudentID = E.StudentID;
```

STUDENTID	NAME	COURSEID
5	Ravi Kumar	309
9	Akriti Sharma	308
16	Rahul Kumar	304
17	Priya Kumari	307
21	Manish Kumar	301
25	Binit Kumar	306
32	Sushil Sagar	302
33	Vivek Kumar	303
38	Prem Kumar	305
51	Raju Kumar	310

10 rows selected.

Q2

```
SQL> SELECT
  2   S.Name,
  3   E.Semester
  4 FROM Student S
  5 INNER JOIN Enrollment E
  6 ON S.StudentID = E.StudentID;
```

NAME	SEMESTER
Ravi Kumar	3
Akriti Sharma	3
Rahul Kumar	5
Priya Kumari	5
Manish Kumar	4
Binit Kumar	5
Sushil Sagar	4
Vivek Kumar	4
Prem Kumar	5
Raju Kumar	4

10 rows selected.

Q3

```
SQL> SELECT
  2   S.Name,
  3   E.Grade
  4 FROM Student S
  5 INNER JOIN Enrollment E
```

6 ON S.StudentID = E.StudentID;

NAME	GRADE
-----	-----
Manish Kumar	A
Sushil Sagar	B+
Vivek Kumar	A
Rahul Kumar	B
Prem Kumar	A
Binit Kumar	A+
Priya Kumari	B+
Akriti Sharma	A
Ravi Kumar	B
Raju Kumar	A

10 rows selected.

Part B: Course–Faculty Relationship

Q4

```
SQL> SELECT
2   C.CourseName,
3   C.Credits,
4   F.Name AS Faculty_Name
5 FROM Course C
6 INNER JOIN Faculty F
7 ON C.FacultyID = F.FacultyID;
```

COURSENAME	CREDITS
-----	-----
FACULTY_NAME	

Database Management System	4
Mr. Sujeet Kumar	
Operating System	4
Mr. Amit Kumar	
Data Structures	4
Mr. Sujeet Kumar	

COURSENAME	CREDITS
-----	-----
FACULTY_NAME	

Structural Analysis	4
Mr. Kunal Kumar	
Engineering Mechanics	3
Mr. Akhilesh Kumar	
Thermodynamics	4
Dr. Ajay Giri	

COURSENAME	CREDITS
------------	---------

FACULTY_NAME	

Strength of Materials	4
Mr. Uttam Kumar	

Basic Electrical Engineering	3
Mr. H. K. Mishra	

Power Systems	4
Ms. Mitesh Kumar	

COURSENAME	CREDITS
------------	---------

FACULTY_NAME	

Machine Learning Fundamentals	4
Md. Haque	

10 rows selected.

Q5

```
SQL> SELECT
  2   C.CourseName
  3 FROM Course C
  4 INNER JOIN Faculty F
  5 ON C.FacultyID = F.FacultyID
  6 WHERE F.FacultyID = 201;
```

COURSENAME

Database Management System
Data Structures

Q6

```
SQL> SELECT
  2   F.Name,
  3   F.Designation,
  4   C.CourseName
  5 FROM Faculty F
  6 INNER JOIN Course C
  7 ON F.FacultyID = C.FacultyID;
```

NAME

DESIGNATION

COURSENAME

Mr. Sujeet Kumar
Assistant Professor

Database Management System

Mr. Amit Kumar
Assistant Professor
Operating System

NAME

DESIGNATION

COURSENAME

Mr. Sujeet Kumar
Assistant Professor
Data Structures

Mr. Kunal Kumar
Assistant Professor

NAME

DESIGNATION

COURSENAME

Structural Analysis

Mr. Akhilesh Kumar
Assistant Professor
Engineering Mechanics

Dr. Ajay Giri

NAME

DESIGNATION

COURSENAME

Assistant Professor
Thermodynamics

Mr. Uttam Kumar
Assistant Professor
Strength of Materials

NAME

DESIGNATION

COURSENAME

Mr. H. K. Mishra

Assistant Professor
Basic Electrical Engineering

Ms. Mitesh Kumar
Assistant Professor
Power Systems

NAME

DESIGNATION

COURSENAME

Md. Haque
Assistant Professor
Machine Learning Fundamentals

10 rows selected.

Part C: Student–Course–Enrollment Relationship

Q7
SQL> SELECT
2 S.Name,
3 C.CourseName
4 FROM Student S
5 INNER JOIN Enrollment E
6 ON S.StudentID = E.StudentID
7 INNER JOIN Course C
8 ON E.CourseID = C.CourseID;

NAME

COURSENAME

Ravi Kumar
Power Systems

Akriti Sharma
Basic Electrical Engineering

Rahul Kumar
Structural Analysis

NAME

COURSENAME

Priya Kumari
Strength of Materials

Manish Kumar

Database Management System

Binit Kumar
Thermodynamics

NAME

COURSENAME

Sushil Sagar
Operating System

Vivek Kumar
Data Structures

Prem Kumar
Engineering Mechanics

NAME

COURSENAME

Raju Kumar
Machine Learning Fundamentals

10 rows selected.

Q8
SQL> SELECT
2 S.Name,
3 C.CourseName,
4 E.Semester
5 FROM Student S
6 INNER JOIN Enrollment E
7 ON S.StudentID = E.StudentID
8 INNER JOIN Course C
9 ON E.CourseID = C.CourseID;

NAME

COURSENAME

SEMESTER

Ravi Kumar
Power Systems 3

Akriti Sharma
Basic Electrical Engineering 3

Rahul Kumar
Structural Analysis 5

NAME	

COURSENAME	SEMESTER
-----	-----
Priya Kumari	
Strength of Materials	5
Manish Kumar	
Database Management System	4
Binit Kumar	
Thermodynamics	5

NAME	

COURSENAME	SEMESTER
-----	-----
Sushil Sagar	
Operating System	4
Vivek Kumar	
Data Structures	4
Prem Kumar	
Engineering Mechanics	5

NAME	

COURSENAME	SEMESTER
-----	-----
Raju Kumar	
Machine Learning Fundamentals	4

10 rows selected.

Q9
 SQL> SELECT
 2 S.Name,
 3 C.CourseName,
 4 E.Grade
 5 FROM Student S
 6 INNER JOIN Enrollment E
 7 ON S.StudentID = E.StudentID
 8 INNER JOIN Course C
 9 ON E.CourseID = C.CourseID
 10 WHERE E.Semester = '4';

NAME	

COURSENAME	GRADE
-----	-----
Manish Kumar	

Database Management System A

Sushil Sagar
Operating System B+

Vivek Kumar
Data Structures A

NAME

COURSENAME GRADE

Raju Kumar
Machine Learning Fundamentals A

Q10
SQL> SELECT
2 C.CourseName
3 FROM Student S
4 INNER JOIN Enrollment E
5 ON S.StudentID = E.StudentID
6 INNER JOIN Course C
7 ON E.CourseID = C.CourseID
8 WHERE S.StudentID = 21;

COURSENAME

Database Management System

Part D: Department-Based JOIN Queries

Q11
SQL> SELECT
2 S.Name,
3 S.DepartmentID,
4 C.CourseName
5 FROM Student S
6 INNER JOIN Enrollment E
7 ON S.StudentID = E.StudentID
8 INNER JOIN Course C
9 ON E.CourseID = C.CourseID;

NAME DEPARTMENTID

COURSENAME

Ravi Kumar 110
Power Systems

Akriti Sharma 157
Basic Electrical Engineering

Rahul Kumar 102
Structural Analysis

NAME	DEPARTMENTID
Priya Kumari	110
Strength of Materials	
Manish Kumar	105
Database Management System	
Binit Kumar	157
Thermodynamics	

NAME	DEPARTMENTID
Sushil Sagar	105
Operating System	
Vivek Kumar	105
Data Structures	
Prem Kumar	105
Engineering Mechanics	

NAME	DEPARTMENTID
Raju Kumar	102
Machine Learning Fundamentals	

10 rows selected.

Q12
 SQL> SELECT
 2 D.DepartmentName,
 3 C.CourseName
 4 FROM Department D
 5 INNER JOIN Course C
 6 ON D.DepartmentID = C.DepartmentID;

DEPARTMENTNAME
CSE
Database Management System

CSE
Operating System

CSE
Data Structures

DEPARTMENTNAME

COURSENAME

CE
Structural Analysis

CE
Engineering Mechanics

ME
Thermodynamics

DEPARTMENTNAME

COURSENAME

ME
Strength of Materials

EEE
Basic Electrical Engineering

EEE
Power Systems

DEPARTMENTNAME

COURSENAME

AIML
Machine Learning Fundamentals

10 rows selected.

Q13
SQL> SELECT
2 F.Name,
3 D.DepartmentName
4 FROM Faculty F
5 INNER JOIN Department D
6 ON F.DepartmentID = D.DepartmentID;

NAME

DEPARTMENTNAME

Mr. Sujeet Kumar
CSE

Mr. Amit Kumar
CSE

Mr. Kunal Kumar
CE

NAME

DEPARTMENTNAME

Mr. Akhilesh Kumar
CE

Dr. Ajay Giri
ME

Mr. Uttam Kumar
ME

NAME

DEPARTMENTNAME

Mr. H. K. Mishra
EEE

Ms. Mitesh Kumar
EEE

Md. Haque
AIML

NAME

DEPARTMENTNAME

Mr. Praveen Kumar
AIML

10 rows selected.

Part E: JOIN with Filtering and Sorting

Q14
SQL> SELECT
2 S.Name,

```
3    C.CourseName
4 FROM Student S
5 INNER JOIN Enrollment E
6 ON S.StudentID = E.StudentID
7 INNER JOIN Course C
8 ON E.CourseID = C.CourseID
9 WHERE S.DepartmentID = 105;
```

NAME

COURSENAME

Manish Kumar
Database Management System

Sushil Sagar
Operating System

Vivek Kumar
Data Structures

NAME

COURSENAME

Prem Kumar
Engineering Mechanics

Q15
SQL> SELECT
2 S.Name,
3 C.CourseName
4 FROM Student S
5 INNER JOIN Enrollment E
6 ON S.StudentID = E.StudentID
7 INNER JOIN Course C
8 ON E.CourseID = C.CourseID
9 ORDER BY S.Name ASC;

NAME

COURSENAME

Akriti Sharma
Basic Electrical Engineering

Binit Kumar
Thermodynamics

Manish Kumar
Database Management System

NAME

COURSENAME

Prem Kumar
Engineering Mechanics

Priya Kumari
Strength of Materials

Rahul Kumar
Structural Analysis

NAME

COURSENAME

Raju Kumar
Machine Learning Fundamentals

Ravi Kumar
Power Systems

Sushil Sagar
Operating System

NAME

COURSENAME

Vivek Kumar
Data Structures

10 rows selected.

Q16
SQL> SELECT
2 F.Name,
3 C.CourseName,
4 C.Credits
5 FROM Faculty F
6 INNER JOIN Course C
7 ON F.FacultyID = C.FacultyID
8 ORDER BY C.Credits DESC;

NAME -----	COURSENAME -----	CREDITS
Ms. Mitesh Kumar	Power Systems	4

Mr. Amit Kumar	
Operating System	4

Mr. Sujeet Kumar	
Data Structures	4

NAME

COURSENAME	CREDITS
------------	---------

Mr. Kunal Kumar	
Structural Analysis	4

Md. Haque	
Machine Learning Fundamentals	4

Dr. Ajay Giri	
Thermodynamics	4

NAME

COURSENAME	CREDITS
------------	---------

Mr. Uttam Kumar	
Strength of Materials	4

Mr. Sujeet Kumar	
Database Management System	4

Mr. H. K. Mishra	
Basic Electrical Engineering	3

NAME

COURSENAME	CREDITS
------------	---------

Mr. Akhilesh Kumar	
Engineering Mechanics	3

10 rows selected.

Part F: Analytical JOIN Queries

Q17

```
SQL> SELECT
2   C.CourseName,
3   COUNT(E.StudentID) AS Student_Count
4 FROM Course C
5 INNER JOIN Enrollment E
6 ON C.CourseID = E.CourseID
7 GROUP BY C.CourseName;
```

COURSENAME	STUDENT_COUNT
Power Systems	1
Basic Electrical Engineering	1
Structural Analysis	1
Strength of Materials	1
Database Management System	1
Thermodynamics	1
Operating System	1
Data Structures	1
Engineering Mechanics	1
Machine Learning Fundamentals	1

10 rows selected.

Q18

```
SQL> SELECT
  2   F.Name,
  3   COUNT(C.CourseID) AS Course_Count
  4 FROM Faculty F
  5 INNER JOIN Course C
  6 ON F.FacultyID = C.FacultyID
  7 GROUP BY F.Name;
```

NAME	COURSE_COUNT
Mr. Sujeet Kumar	2
Mr. Amit Kumar	1
Mr. Kunal Kumar	1
Mr. Akhilesh Kumar	1
Dr. Ajay Giri	1
Mr. Uttam Kumar	1
Mr. H. K. Mishra	1
Ms. Mitesh Kumar	1
Md. Haque	1

9 rows selected.

Q19

```
SQL> SELECT
  2   D.DepartmentName,
  3   COUNT(S.StudentID) AS Student_Count
  4 FROM Department D
  5 INNER JOIN Student S
  6 ON D.DepartmentID = S.DepartmentID
  7 GROUP BY D.DepartmentName;
```

DEPARTMENTNAME	STUDENT_COUNT
CSE	4
ME	2
AIML	2
EEE	2

Q20

```
SQL> SELECT
2   C.CourseName,
3   MAX(E.Grade) AS Max_Grade
4 FROM Course C
5 INNER JOIN Enrollment E
6 ON C.CourseID = E.CourseID
7 GROUP BY C.CourseName;
```

COURSENAME	MAX_G
Database Management System	A
Operating System	B+
Data Structures	A
Structural Analysis	B
Engineering Mechanics	A
Thermodynamics	A+
Strength of Materials	B+
Basic Electrical Engineering	A
Power Systems	B
Machine Learning Fundamentals	A

10 rows selected.